

Lecture 4

Matrices of Linear Maps; Change of Basis

Throughout, bases are ordered and $\mathcal{M}(T)$ denotes the matrix of T with respect to the bases in force. Tags indicate the flavour of each problem; a ★ marks a harder one.

Core tutorial route

Work **4.1, 4.7, 4.11, 4.14, 4.15, 4.16, 4.17, and 4.20** in that order (about 60–75 minutes before discussion). This eight-problem route checks matrix construction, the composition argument, invertibility, map-space dimension, finite isomorphism structure, a full similarity calculation, coordinate reconstruction, and an invariant proof.

Additional practice: 4.3–4.4, 4.8–4.10, 4.12–4.13, 4.18–4.19. **Bridge:** 4.2, 4.5, 4.6 (polynomial calculus). **Extension:** 4.21–4.23.

Matrices of maps

Exercise 4.1 (computational) Let $T \in \mathcal{L}(\mathbb{R}^3, \mathbb{R}^2)$ be $T(x, y, z) = (x + 2y - z, 3x + z)$. Write $\mathcal{M}(T)$ with respect to the standard bases, and use it to compute $[T(1, 1, 1)]$.

Answer. $\mathcal{M}(T) = \begin{pmatrix} 1 & 2 & -1 \\ 3 & 0 & 1 \end{pmatrix}$; $[T(1, 1, 1)] = (2, 4)$.

Exercise 4.2 (computational) Let $D \in \mathcal{L}(\mathcal{P}_2(\mathbb{R}))$ be differentiation. Compute $\mathcal{M}(D)$ and $\mathcal{M}(D)^2$ in the standard basis $1, t, t^2$, and identify the operator that $\mathcal{M}(D)^2$ represents.

Answer. $\mathcal{M}(D) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$, $\mathcal{M}(D)^2 = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \mathcal{M}(D^2)$.

Exercise 4.3 (computational) Let $T \in \mathcal{L}(\mathbb{R}^2)$ be $T(x, y) = (2x - y, x + 3y)$. Write $\mathcal{M}(T)$ in the standard basis and use it to compute $[T(1, 2)]$.

Answer. $\mathcal{M}(T) = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$; $[T(1, 2)] = (0, 7)$.

Exercise 4.4 (computational) Let $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^3)$ be $T(x, y) = (x - y, 2x + y, 4y)$. Write $\mathcal{M}(T)$ with respect to the standard bases and compute $[T(3, -1)]$.

Answer. $\mathcal{M}(T) = \begin{pmatrix} 1 & -1 \\ 2 & 1 \\ 0 & 4 \end{pmatrix}$; $[T(3, -1)] = (4, 5, -4)$.

Exercise 4.5 (computational) Let $T \in \mathcal{L}(\mathcal{P}_2(\mathbb{R}))$ be $(Tp)(t) = p(t) + t p'(t)$. Compute $\mathcal{M}(T)$ in the standard basis $1, t, t^2$ and use it to evaluate $T(2 - t + 3t^2)$.

Answer. $\mathcal{M}(T) = \text{diag}(1, 2, 3)$; $T(2 - t + 3t^2) = 2 - 2t + 9t^2$.

Exercise 4.6 (computational) Let $D \in \mathcal{L}(\mathcal{P}_3(\mathbb{R}))$ be differentiation, with $\mathcal{M}(D) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ in the standard basis $1, t, t^2, t^3$. Compute $\mathcal{M}(D)^3$ and find the smallest k with $D^k = 0$.

Answer. $\mathcal{M}(D)^3 = \begin{pmatrix} 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$; smallest k is $k = 4$.

Exercise 4.7 (proof) Suppose $S, T \in \mathcal{L}(V)$ where $\dim V = n$, with matrices $\mathcal{M}(S), \mathcal{M}(T)$ in a fixed basis. First derive $[STv] = \mathcal{M}(S)\mathcal{M}(T)[v]$ for an arbitrary v and explain why

this forces $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)$. Then prove that $ST = TS$ as operators if and only if $\mathcal{M}(S)\mathcal{M}(T) = \mathcal{M}(T)\mathcal{M}(S)$ as matrices.

Exercise 4.8 (proof) Fix bases of V and W . Prove that for $S, T \in \mathcal{L}(V, W)$ and $a \in \mathbb{F}$, $\mathcal{M}(aS + T) = a\mathcal{M}(S) + \mathcal{M}(T)$.

Invertibility and isomorphism

Exercise 4.9 (computational) Decide whether $T(x, y) = (2x + y, 4x + 2y)$ on $\mathbb{R}^2$ is invertible. If so give T^{-1} ; if not give a nonzero vector in $\text{null } T$.

Answer. Not invertible ($\det = 0$); $(1, -2) \in \text{null } T$.

Exercise 4.10 (computational) Show that $1, 1 - t, (1 - t)^2$ is a basis of $\mathcal{P}_2(\mathbb{R})$ by transporting the question to $\mathbb{R}^3$.

Answer. Yes — the coordinate vectors $(1, 0, 0), (1, -1, 0), (1, -2, 1)$ are independent ($\det = -1$), so it is a basis.

Exercise 4.11 (computational) Decide whether $T(x, y) = (3x + 5y, x + 2y)$ on $\mathbb{R}^2$ is invertible; if so, give T^{-1} explicitly.

Answer. Invertible ($\det = 1$); $T^{-1}(x, y) = (2x - 5y, -x + 3y)$.

Exercise 4.12 (computational) Decide whether $T(x, y, z) = (x + 2y + 3z, 2x + 4y + 6z, x + y + z)$ on $\mathbb{R}^3$ is invertible. If not, give a nonzero vector in $\text{null } T$.

Answer. Not invertible ($\det = 0$); $(1, -2, 1) \in \text{null } T$.

Exercise 4.13 (computational) Show that $1 + t^2, t + t^3, 1 + t, t^2$ is a basis of $\mathcal{P}_3(\mathbb{R})$ by transporting the question to $\mathbb{R}^4$.

Answer. Yes — the coordinate vectors have determinant $1 \neq 0$, so they are independent, hence a basis.

Exercise 4.14 (proof) Prove that if V and W are finite-dimensional, then $\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$.

Exercise 4.15 (proof) Let V be finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the following are equivalent: (i) T is injective; (ii) T is surjective; (iii) T is invertible.

Hint. Use rank-nullity: $\dim \text{null } T + \dim \text{range } T = \dim V$.

Change of basis and similarity

Exercise 4.16 (computational) Let $T(x, y) = (x, -y)$ on $\mathbb{R}^2$ (reflection in the x -axis), with standard matrix $A = \text{diag}(1, -1)$. Find its matrix in the basis $\mathcal{B}' = ((1, 1), (1, -1))$ using $A' = P^{-1}AP$.

Answer. $A' = P^{-1}AP = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Exercise 4.17 (computational) In $\mathbb{R}^2$ let $\mathcal{B}$ be the standard basis and $\mathcal{B}' = ((2, 1), (1, 1))$. A vector has new coordinates $[v]_{\mathcal{B}'} = (1, 2)$; find $[v]_{\mathcal{B}}$ (i.e. v itself). Also find the $\mathcal{B}'$ -coordinates of $w = (3, 0)$.

Answer. $[v]_{\mathcal{B}} = (4, 3)$; $[w]_{\mathcal{B}'} = (3, -3)$.

Exercise 4.18 (computational) Let $T \in \mathcal{L}(\mathbb{R}^2)$ have standard matrix $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$. Find its matrix in the basis $\mathcal{B}' = ((1, 0), (1, 1))$ using $A' = P^{-1}AP$.

Answer. $A' = \text{diag}(2, 3)$.

Exercise 4.19 (computational) Using trace and determinant only, decide whether each pair of matrices can possibly be similar: (a) $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$; (b) $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ and $\begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}$.

Answer. (a) No (traces $5 \neq 4$). (b) Not ruled out (both have trace 4, determinant 4).

Exercise 4.20 (proof) Prove that similar matrices have the same determinant and the same trace.

Exercise 4.21 (★) Suppose A is similar to B and B is similar to C . Prove A is similar to C , and deduce that being similar is an equivalence relation on $n \times n$ matrices.

Hint. For transitivity, compose the two conjugating matrices and simplify $(PQ)^{-1}A(PQ)$.

Exercise 4.22 (★) Suppose A and B are $n \times n$ matrices with A invertible. Prove that AB and BA are similar, and conclude $\text{tr}(AB) = \text{tr}(BA)$.

Hint. You need a single matrix P with $P^{-1}(AB)P = BA$. Since A is invertible, try $P = A$.

Exercise 4.23 (★) Give two 2×2 matrices with the same trace and the same determinant that are nevertheless *not* similar, and prove they are not similar.

Hint. Compare $3I$ with a non-diagonal matrix of the same trace and determinant. What is $P^{-1}(3I)P$?

Answer. E.g. $\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$ and $\begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$: both have trace 6 and determinant 9, but are not similar.